

# The Principle of Inclusion-Exclusion

**Theorem.** Let  $(\Omega, \mathcal{S}, P)$  be a probability space, and let  $A_1, A_2, \dots, A_n \in \mathcal{S}$ ,  $n \geq 1$ . Then:

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i_1=1}^n P(A_{i_1}) - \sum_{1 \leq i_1 < i_2 \leq n} P(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{n-1} P(A_1 \cap \dots \cap A_n) \\ &= \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P(A_{i_1} \cap \dots \cap A_{i_k}). \end{aligned}$$

In other words, the probability of the union of events equals the sum of individual probabilities, minus the sum of pairwise intersection probabilities, plus the sum of triple intersection probabilities, and so forth, with signs alternating.

**Proof by induction on  $n$ .**

*Base cases:* For  $n = 1$ , the formula reduces to  $P(A_1) = P(A_1)$ , which holds by the definition of equality. For  $n = 2$ , we showed in class that  $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$ .

*Inductive step:* Assume the theorem holds for any collection of  $n - 1$  events. We shall prove it for a collection  $A_1, \dots, A_n$  of  $n$  events. Write

$$\bigcup_{i=1}^n A_i = \left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n.$$

Let  $B = \bigcup_{i=1}^{n-1} A_i$ . By the inclusion-exclusion formula for two events, we get:

$$P(B \cup A_n) = P(B) + P(A_n) - P(B \cap A_n).$$

Define  $C_i := A_i \cap A_n$  for each  $1 \leq i \leq n - 1$ . (Note that each  $C_i$  is also an event.) Now, we observe that:

$$B \cap A_n = \left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n = \bigcup_{i=1}^{n-1} (A_i \cap A_n) = \bigcup_{i=1}^{n-1} C_i.$$

By the induction hypothesis applied to the events  $A_1, \dots, A_{n-1}$  for computing  $P(B)$ , and to the events  $C_1, \dots, C_{n-1}$  for computing  $P(B \cap A_n)$ , we get:

$$\begin{aligned} P(B) &= \sum_{k=1}^{n-1} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} P(A_{i_1} \cap \dots \cap A_{i_k}), \\ P(B \cap A_n) &= \sum_{k=1}^{n-1} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} P(C_{i_1} \cap \dots \cap C_{i_k}) \\ &= \sum_{k=1}^{n-1} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} P(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n), \end{aligned}$$

where the last equality holds because

$$C_{i_1} \cap \cdots \cap C_{i_k} = (A_{i_1} \cap A_n) \cap \cdots \cap (A_{i_k} \cap A_n) = A_{i_1} \cap \cdots \cap A_{i_k} \cap A_n.$$

Substituting into  $P(B \cup A_n) = P(B) + P(A_n) - P(B \cap A_n)$ , we get:

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k-1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} P(A_{i_1} \cap \cdots \cap A_{i_k}) \\ &\quad + P(A_n) - \sum_{k=1}^{n-1} (-1)^{k-1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} P(A_{i_1} \cap \cdots \cap A_{i_k} \cap A_n). \end{aligned}$$

Now we are almost done. The RHS has the terms  $P(A_1) + \cdots + P(A_{n-1})$  from the  $k = 1$  term in the first sum. This, along with the  $P(A_n)$  term, gives  $(-1)^0 \sum_{1 \leq i_1 \leq n} P(A_{i_1})$  as required.

Next, the first sum on the RHS for a fixed  $k \geq 2$  contains all the terms  $P(A_{i_1} \cap \cdots \cap A_{i_k})$  such that  $i_n \notin \{i_1, \dots, i_k\}$ , and the last sum on the RHS at  $k - 1 \geq 1$  contains all the terms  $P(A_{i_1} \cap \cdots \cap A_{i_{k-1}} \cap A_{i_n})$ . Moreover, these all have the correct sign,  $(-1)^k$  (check this carefully!). Thus, we get:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} P(A_{i_1} \cap \cdots \cap A_{i_k}).$$

This completes the proof by induction. □